

Zhihan Rachel Hua

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SUMMARY

Ph.D.-trained computational biologist with extensive experience developing AI-enabled bioinformatics workflows for genomics and multi-omics research. Specialized in designing scalable analysis pipelines for spatial transcriptomics, single-cell sequencing, long-read genomics, and epigenomics, with a strong emphasis on reproducibility, automation, and biological interpretation. Experienced working across research and product teams to build computational tools that accelerate scientific discovery and support translational research.

SKILLS

Programming & Data Science: Python, R, SQL, Bash, Unix/Linux, Shell Scripting, Pandas, NumPy, SciPy, scikit-learn, Jupyter, VS Code
Single-cell & Spatial Genomics: Scanpy, Seurat, Squidpy, AnnData, CellTypist, spatial transcriptomics (10x Xenium, Visium, CODEX/PhenoCycler), spatial statistics, neighborhood/niche analysis, cell-type annotation, ligand-receptor analysis, multi-omic integration (Harmony, MaxFuse, uniPort)

Genomics & Bioinformatics: NGS, RNA-seq (DESeq2, edgeR, limma), single-cell RNA-seq, pseudobulk analysis, pathway enrichment (GSEA, fgsea, CAMERA), long-read sequencing (Oxford Nanopore, PacBio), variant calling (GATK, BCFtools, samtools), alignment (BWA, Bowtie2, minimap2), GWAS (PLINK), genome assembly & QC (SPAdes, QUAST), variant annotation (snpEff), epigenomics (CUT&Tag, MACS3, deepTools), visualization (IGV)

Machine Learning & AI: AI-assisted bioinformatics workflows, LLM applications for scientific analysis, retrieval-augmented generation (RAG), statistical modeling, clustering, dimensionality reduction (PCA, UMAP, t-SNE), model benchmarking

Workflow Automation & Infrastructure: Nextflow, Snakemake, Docker, HPC (SLURM), AWS, Google Cloud Platform (GCP), Lambda Cloud, Vercel, GitHub Actions, pipeline development, cloud deployment, reproducible workflows

Version Control & Software Engineering: Git, GitHub, CI/CD, Docker, code review, software documentation, API integration

RESEARCH EXPERIENCE

Bioinformatics Engineer April 2026 – Present

Novaflow (YC-backed bioinformatics platform) | Los Angeles, CA

- Lead bioinformatics analyses across 20+ client projects spanning academia, biotechnology, and pharmaceutical research, delivering interactive web portals with reproducible workflows, downloadable datasets, publication-quality figures, and technical reports.
- Analyze **multi-omics** datasets including **spatial transcriptomics** and **proteomics** (10x Xenium, CODEX/PhenoCycler), **single-cell RNA-seq**, **NGS**, **long-read** sequencing (Oxford Nanopore), **epigenomics** (CUT&Tag), and **translational profiling** (Ribo-seq/Polysome-seq) to generate biologically meaningful insights.
- Used the **1000 Genomes Project** to characterize population-specific SNP variation and allele-frequency patterns, investigating variants associated with **pathogenic repeat expansions** and estimating candidate carrier frequencies across diverse global populations.
- Apply AlphaFold, ESMFold, and RFdiffusion for **protein structure prediction** and structure-guided **antibody optimization**, using in silico mutagenesis and binding/stability assessment to prioritize Ephrin-B2/EphB4 variants with improved target binding.
- Analyzed **single-cell RNA-seq** from the PURE01 muscle-invasive bladder cancer trial, processing raw FASTQs with STARsolo, performing cell-type annotation and CNV-based malignant-cell identification, and profiling the tumor immune microenvironment and **EphB4-ephrin-B2 therapeutic axis** to distinguish malignant from benign urothelium and improve tumor subtype classification.
- Analyzed H3K27me3/H3K9me2 **CUT&Tag** in WT and Eed-knockout mouse trophoblast stem cells, developing a reproducible QC pipeline to assess alignment, spike-in normalization, antibody enrichment, peak/domain structure, and IP-IgG signal; identified assay-wide enrichment failure and over-tagmentation, preventing misleading downstream differential analysis and guiding experimental redesign.
- Analyzed **polysome profiling RNA-seq** in **CR31B-treated U937 acute myeloid leukemia cells**, developing an end-to-end workflow for sequencing QC, alignment, differential translation efficiency and polysome-to-monosome analysis, and 5'UTR characterization to identify sequence and structural features, including G-quadruplexes, RNA folding, and motifs-associated with translational regulation.
- Developed an **Oxford Nanopore long-read pipeline** for B6/FVB ancestry mapping and loss-of-heterozygosity (LOH) breakpoint detection, using dual-reference competitive alignment, per-read ancestry inference, and HMM segmentation to recover 27 high/medium-confidence transitions and characterize genome-wide LOH patterns in a diploid mouse cell line.
- Integrated **10x Xenium spatial transcriptomics with matched H&E histology** for a 218K+ cell Merkel cell carcinoma tissue microarray, developing image-registration and spatial analysis workflows to align molecular and histological features, segment tissue cores, identify tumor regions, and characterize tumor-immune microenvironment heterogeneity.
- Integrated **10x Xenium spatial transcriptomics with CODEX/PhenoCycler spatial proteomics** across five Parkinson's disease mouse genotypes to characterize genotype-specific changes in gene and protein expression, cell-type composition and states,

spatial organization, and multicellular tissue niches, linking molecular alterations to disease-associated tissue phenotypes.

- Developed **spTranslation**, a unified benchmark for spatial transcriptomics translation across histology-to-expression, super-resolution, scRNA-to-spatial imputation, and cross-platform integration, benchmarking state-of-the-art methods across 11 datasets using expression fidelity and spatial-structure metrics to identify methods that best preserve biologically meaningful spatial patterns.
- Built **BrainLike**, an interactive benchmarking portal for iPSC-derived brain-cell models, comparing microglia and neurons against experimental-floor and native-tissue-ceiling references by integrating **11 published datasets across 10 public resources**, including GEO, SRA, Allen Brain Cell Atlas, CELLxGENE, Human Protein Atlas, UniProt, STRING, and HCP.
- Mentor junior bioinformaticians and interns by reviewing code, analysis pipelines, and experimental designs, while contributing to internal method development and platform improvements.

Postdoctoral Fellow Sept 2025 – April 2026

Ambry Genetics Tempus AI | Aliso Viejo, CA

- Executed and optimized PacBio long-read sequencing workflows, including sample selection, DNA extraction, library preparation, and QC, enabling consistent, high-quality data generation on a weekly basis and driving the development of a scalable, productized sequencing pipeline.
- Partnered with scientists, automation engineers, and operations teams to design and optimize **Tecan and Hamilton liquid-handling workflows** for scalable, high-throughput sequencing operations.
- Built downstream **NGS pipelines** to identify structural variants and genome-wide features from long-read datasets.
- Interpreted and curated rare-disease variants using **ACMG criteria and population/clinical databases including gnomAD and ClinVar**, integrating allele frequencies, clinical annotations, and PacBio long-read evidence to assess variant pathogenicity and present findings in weekly cross-functional reviews.
- Contributed to process documentation, workflow standardization, and system optimization, supporting reproducibility, compliance, and knowledge transfer across teams. Evaluated and recommended improvements to scientific data workflows, enhancing data integration, traceability, and usability across systems.

Graduate Student Researcher August 2020 – August 2025

Nuzhdin Lab at USC | Los Angeles, CA

- Transcriptomic Analysis of Organism Development: Spearheaded **RNA-seq expression analysis** in 3-day-old and 7-day-old larvae, identifying developmental pathways and deformity-associated genes. Developed customized R/Python pipelines for transcriptomic and functional genomics analysis, applying statistical methods to large-scale datasets using **HPC**.
- Germline Mutation Detection: Designed and optimized high-throughput **WGS bioinformatics workflows** using Python, R, and Bash to detect de novo germline mutations in bivalves. Implemented variant calling, joint genotyping, and customized QC filtering for depth, allelic balance, and strand bias to reduce false positives and enable robust mutation-rate estimation.
- Machine Learning & Biomarker Discovery: Designed and executed a field-based molecular ecology study of wildfire-related stressors in marine ecosystems, developing predictive machine-learning models to identify molecular biomarkers and differentially expressed genes associated with stress response and immune function. Integrated RNA-seq and environmental metadata to characterize enriched pathways and population-level effects.
- Genomic Selection & Trait Mapping: Applied genomic-assisted breeding approaches to bivalve aquaculture, integrating multi-omics data to identify SNPs associated with growth, survival, and stress resilience. Conducted **GWAS** using mixed linear models and selective-sweep analyses with XP-CLR and FST to identify trait-associated loci and regions under selection.
- Mutagenesis Variant Calling Study: Developed a forward-genetic screening and **variant-calling pipeline** for *Macrocystis pyrifera* gametophytes following EMS mutagenesis, combining phenotypic screening with genomic analysis to map candidate causal mutations associated with motility defects and developmental disruption.

EDUCATION

University of Southern California — Ph.D., Computational and Molecular Biology 2020 – 2025 **University of California, Los Angeles — B.S. in Biochemistry** 2016 – 2020

PUBLICATIONS AND PRESENTATIONS

1. Zhihan Hua, Nathan Churches, Sergey V. Nuzhdin. Balancing selection and candidate loci for survival and growth during larval development in the Mediterranean mussel, *Mytilus galloprovincialis*. *G3 Genes|Genomes|Genetics*, 13(7), July 2023, jkad103.
2. Zhihan Hua, Nathan D. Churches, Jordan Chancellor, Peter L. Chang, Sergey V. Nuzhdin. Pacific oysters (*Crassostrea gigas*) dramatically recalibrate models for the upper limit of the eukaryotic mutation rate (manuscript in review).
3. Zhihan Hua, Sergey Nuzhdin. Population Genomics of Mediterranean Mussels, *Mytilus galloprovincialis*, Uncovers Divergent Selection Signatures Between Core and PAV Gene Families (manuscript in review).
4. Zhihan Hua. A corrected ranking axis for spatial-transcriptomics translation: report $z(\text{PCC}) + z(\text{LISA})$, because PCC-only leaderboards misrank methods for downstream spatial analysis (manuscript in review).